

MUHAMMAD ARIF KHAN

Software Engineer | Backend Development | AI & Automation Systems
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Summary

Software Engineer specializing in backend development, automation systems, and AI-assisted software engineering. Experienced in designing secure REST APIs, authentication systems, workflow automation platforms, and intelligent developer tools using Java, Spring Boot, Python, FastAPI, Laravel, and PostgreSQL. Proven ability to build scalable, maintainable software solutions that solve real-world business and operational challenges. Passionate about combining software engineering, automation, and AI technologies to create impactful products.

Technical Skills

Programming Languages: Java · Python · Kotlin · PHP

Backend Development: Spring Boot · FastAPI · Laravel · REST APIs · JWT Authentication · Spring Security

Databases: MySQL, PostgreSQL

Tools: Git · GitHub · Docker · Postman · Linux

Professional Experience

University of Sharjah | Remote

Research Assistant

2024 - 2025

- Conducted software research and prototype development for experimental and research-oriented systems.
- Designed, implemented, and evaluated software solutions to validate technical feasibility and performance.
- Performed debugging, optimization, and performance analysis on research codebases.
- Produced technical documentation and contributed to research reporting and project evaluation activities.

Iqra University | Karachi, Pakistan

Research Assistant

2022 - 2021

- Coordinated with the IT department to ensure seamless system integrations, resulting in improved data flow and accuracy across platforms.
- Developed backend services and Android-based applications supporting academic research initiatives.
- Designed and integrated REST APIs to improve communication between application components.
- Improved software maintainability through refactoring, optimization, and structured debugging practices.
- Assisted in prototype validation, software testing, and technical documentation.

Projects

CarpeFlow CRM – Business Automation & CRM Platform

<https://github.com/arifxlab/carpeflow-crm>

Developing a CRM and workflow automation platform using Laravel, PHP, PostgreSQL, and Docker.

- Designing customer lifecycle management, lead tracking, role-based access control, and business process automation workflows.
- Building scalable backend architecture to support reporting, automation, and operational efficiency.
- Technologies: Laravel, PHP, PostgreSQL, Docker, REST APIs

Dev Assistant AI – AI-Powered Static Code Analysis Platform

<https://github.com/arifxlab/dev-assistant-ai>

Built an intelligent code analysis platform that evaluates source code quality using AST processing and maintainability metrics.

- Implemented complexity analysis, nesting evaluation, maintainability scoring, and structured reporting APIs.
- Designed scalable architecture for future AI-assisted code review and refactoring capabilities.
- Technologies: Python, FastAPI, AST Processing, JSON APIs

Projects

NeuralOps – Enterprise Backend Architecture Platform

<https://github.com/arifxlab/neuralops-core>

Developed a production-style backend platform emphasizing secure API development, layered architecture, and scalable service design.

- Implemented modular backend components following enterprise software engineering principles.
- Strengthened expertise in backend architecture, API design, and maintainable code practices.
- Technologies: Java, Spring Boot, MySQL, REST APIs

AquaGrid – Authentication & Authorization System

<https://github.com/arifxlab/aquagrid-core>

Designed and implemented a secure authentication platform using Spring Security and JWT.

- Developed user registration, login, password encryption, token-based authentication, and role-based access control.
- Secured application endpoints and implemented best practices for API security.
- Technologies: Java, Spring Boot, Spring Security, JWT, MySQL

Education

Iqra University, Karachi, Pakistan

2022 - 2027

Bachelor of Software Engineering